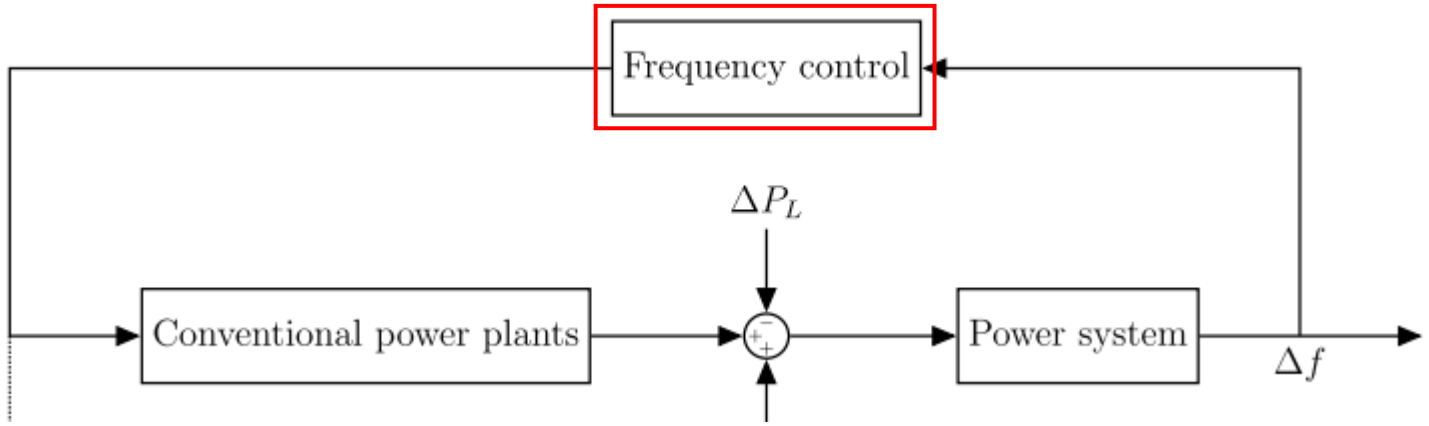
Control Theory and Applications (01)

Frequency Stabilization Control Design for Turbine Systems

Yunui Kang Dongseok Kim Huijeong Choi

Objective



: Maintenance of power quality through frequency control

» generators (synchronous machines) to enable parallel operation
» stable power supply
» preventing damage such as blade breakage and overheating

Turbine System Composition

Single-Reheat Steam Turbine



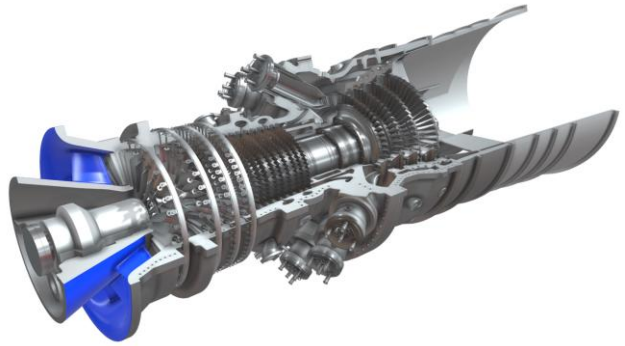
$$P_{sp} \rightarrow \boxed{\frac{1}{1+s\,T_g}} \rightarrow \boxed{\frac{1}{1+s\,T_{CH}}} \rightarrow \boxed{\frac{1+sF_{HP}\,T_{RH}}{1+s\,T_{RH}}} \rightarrow P_g$$

P_{sp} 발전기 출력 목표

P_g 발전기 출력 전력

governor(증기 공급) → HP 터빈 → 재열기 → LP 터빈 → 출력

Gas Turbine



$$P_{sp} \rightarrow \boxed{\frac{a}{c+s\,b}} \rightarrow \boxed{\frac{1+s\,X}{1+s\,Y}} \rightarrow \boxed{\frac{1+s\,T_{CR}}{1+s\,T_F}} \rightarrow \boxed{\frac{1}{1+s\,T_{CD}}} \rightarrow P_{mech}$$

P_{sp} 발전기 출력 목표

P_{mech} 발전기 출력 전력

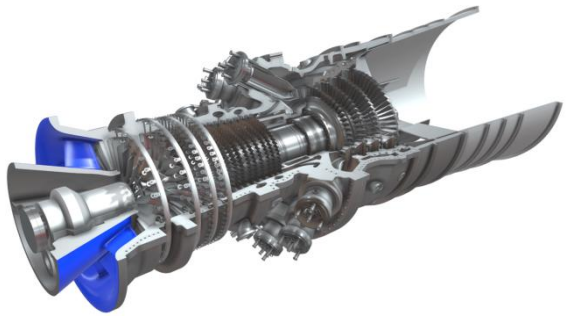
governor(연료밸브 조정) → 연소실 → 터빈 → 에너지 출력

Turbine System Composition

Single-Reheat Steam Turbine



Gas Turbine



Characteristic	High efficiency, slow dynamic response	Fast response, excellent maneuverability
Used for	base	Fast backup
Controlling	Steam	Gas

Turbine System Composition

Single-Reheat Steam Turbine + Gas Turbine

Multiple turbines coexisted in actual systems

Gas turbines are responsible for quick response,
while steam turbines are responsible for stable continuous output.

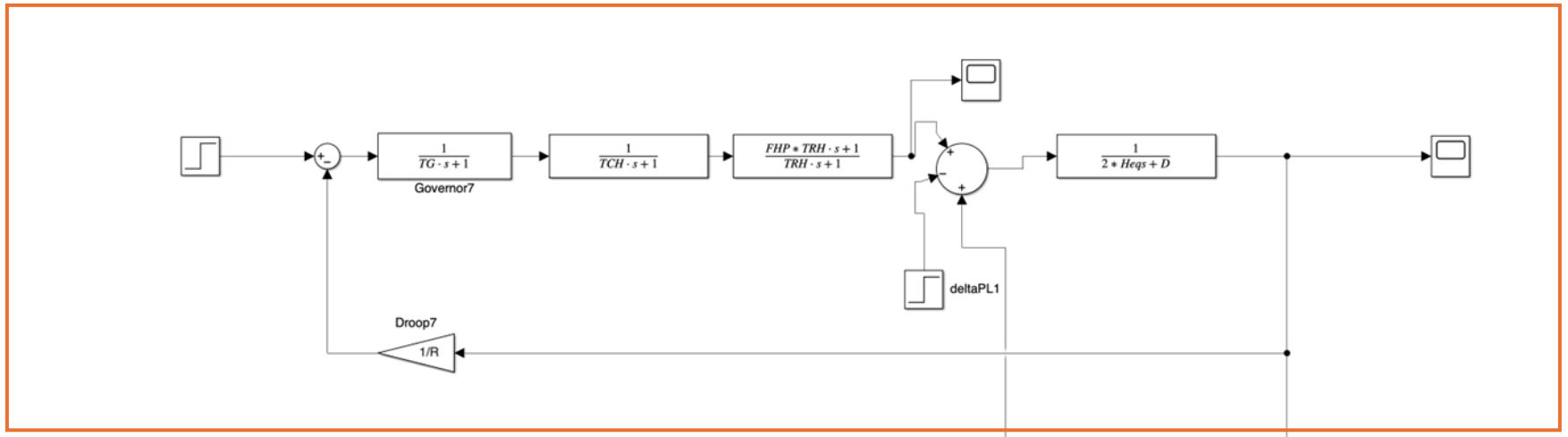
↓

Integrated control of both types of turbines

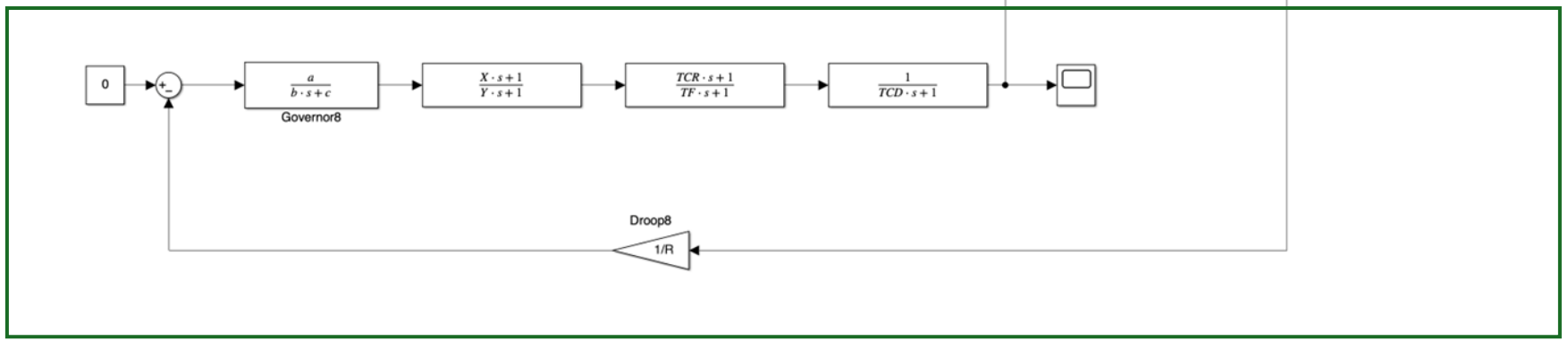
Accurate control design that reflects mutual cooperation..

Turbine System Composition

Single-Reheat Steam Turbine



Gas Turbine



Turbine System Composition



Single reheat steam turbine

Gain margin: 10.7db
Phase margin: 53.2deg



Gas turbine

Gain margin: 11.1db
Phase margin: 38.2deg



Single reheat + Gas turbine

Gain margin: 0.373db
Phase margin: 1.26deg

Method

3-Step Methodology

1

Droop Control

2

Step Input Recover

3

Compensator Design

Method

1. Droop control

: Primary control (fast response)

$$\Delta P = -\frac{1}{R} \cdot \Delta f$$

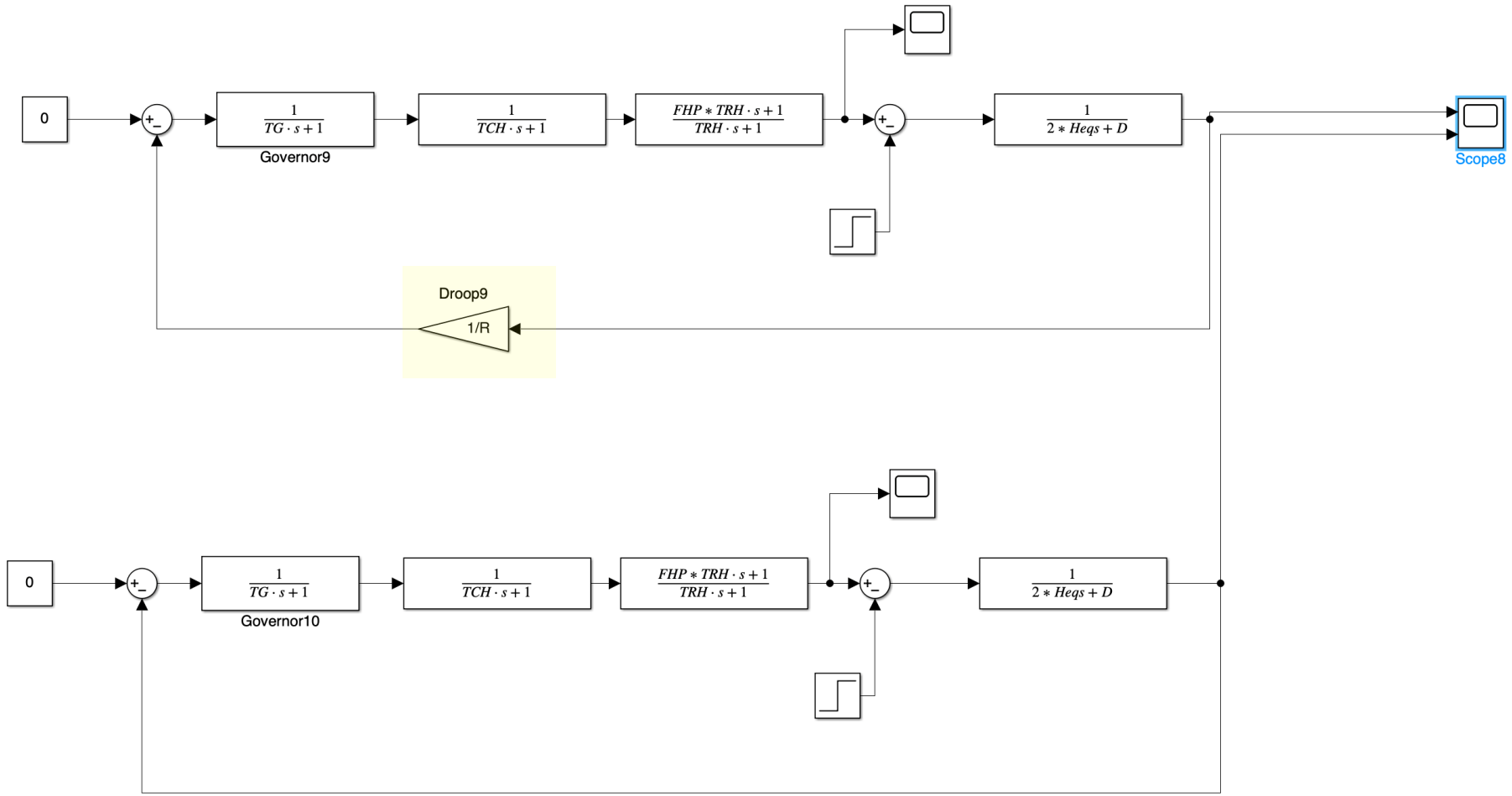
R 드룹 비율
Δf 기준 주파수 대비 편차
ΔP 출력 전력의 변화량

» Automatically changes output power according to frequency changes.
» Multiple generators cooperatively share the load.

Without droop control, there is a risk that each generator will operate at a different frequency.

Method

1. Droop control



Method

1. Droop control

$$R = \frac{\Delta f}{\Delta P} \leftrightarrow \Delta P_{droop} = \frac{1}{R}\Delta f \qquad \boxed{R = 0.05}$$

5% Droop 주파수가 1% 떨어질 때 발전기 출력이 5% 증가하도록 설계

» 부하 분담 균일화

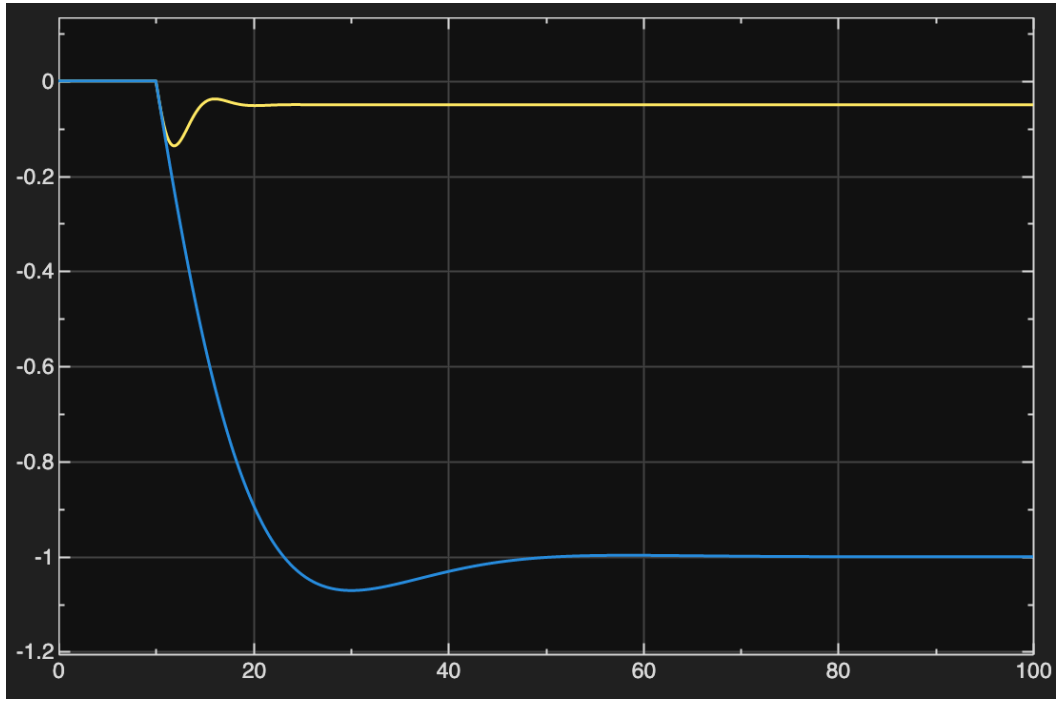
$$\Delta P_1 = \frac{1}{R}\Delta\omega \qquad \Delta P_2 = \frac{1}{R}\Delta\omega$$

: pu 단위 부하 보충량이 일치

» 1차 응답 특성 일관성

Method

1. Droop control



Droop Control Applied Without Droop Control

Input: 부하 스텝 ($\Delta P: 0 \rightarrow 1\ pu\ @\ t = 10s$)

Without Droop Control

$$M\frac{d\Delta\omega}{dt} + D\Delta\omega = 0 - 1 = -1$$ Speed continues to decrease

Since inertia and damping alone do not provide control gain, convergence occurs in a state of **frequency collapse**

Droop Control Applied

$$P_{ref} = \frac{1}{R}\Delta\omega - \Delta P_L$$ 비례제어이득이 걸림

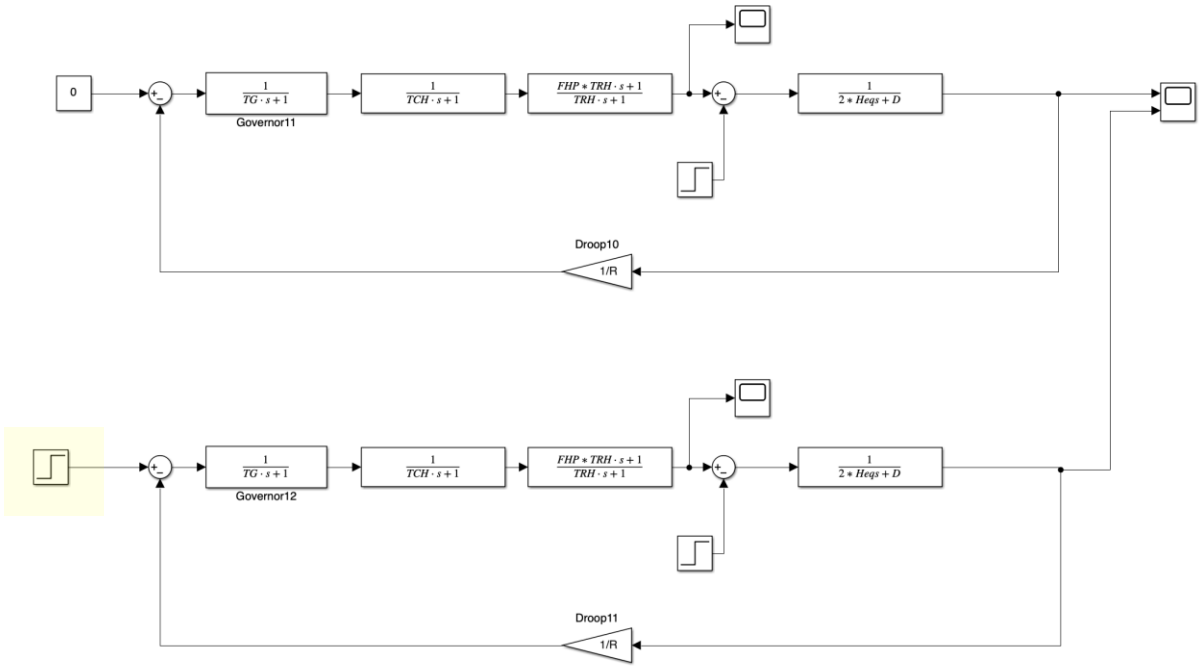
부하 스텝 직후 토크가 $-1 + \frac{1}{R}\Delta\omega$로 바뀌므로 순간 약 -0.15 pu 정도 급락했다가 비례제어이득이 작용해 일부 회복

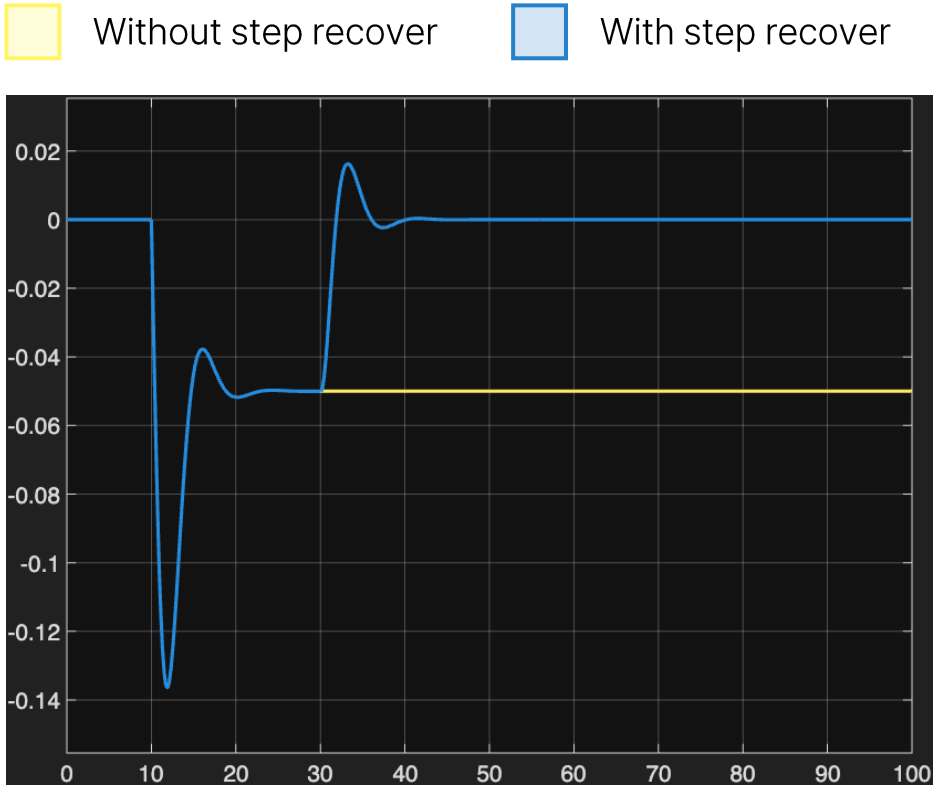
$$\Delta\omega_\infty = -R\Delta P_L = -0.05 * 1 = -0.05\ \text{pu}$$

정상상태 편차 5% 드룹 대로 주파수 드롭이 제한됨

Method

2. Step Input Recover

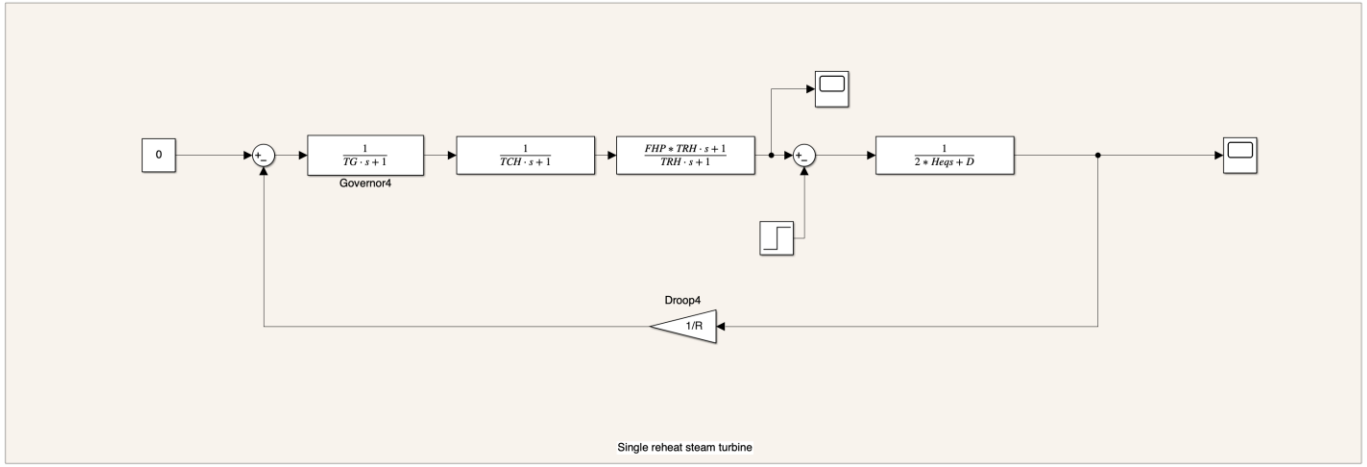




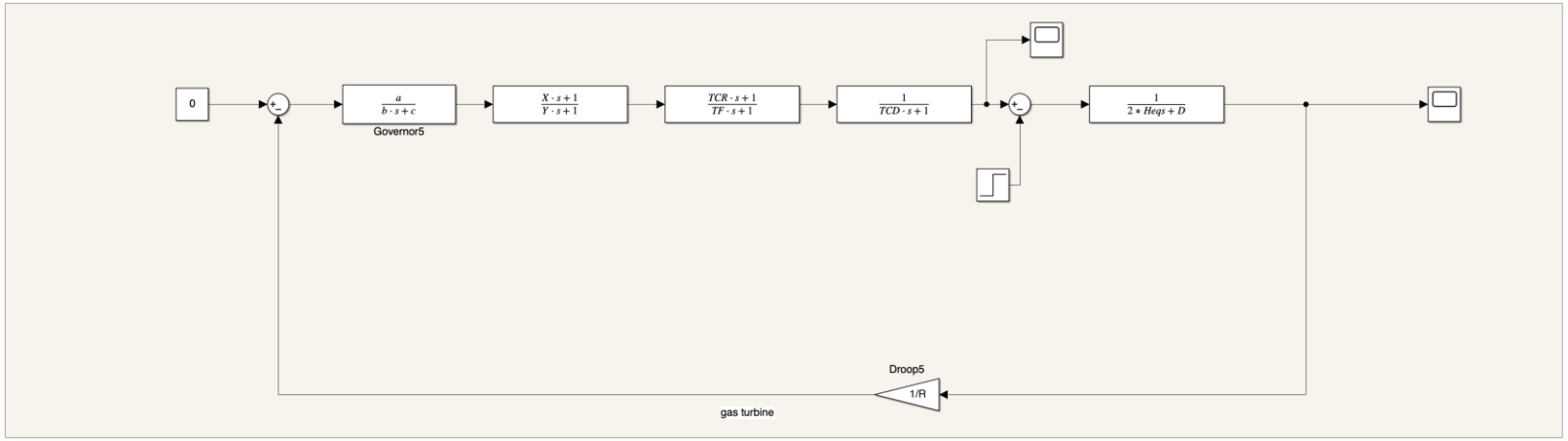
발전기 Governor은 주파수 편차 Δf에 비례해 기계 출력 ΔP_m을 조정
5% droop이면 $\Delta f = -1\%$일 때 $\Delta P = 5\%$를 보충
부하 스텝 시 주파수는 일단 하락했다가 $\Delta f_\infty = -R\Delta P_L = -0.05$ pu 에서 멈춤

→ **Step Input Recover**

3. Compensator Design



◀ Single reheat steam turbine



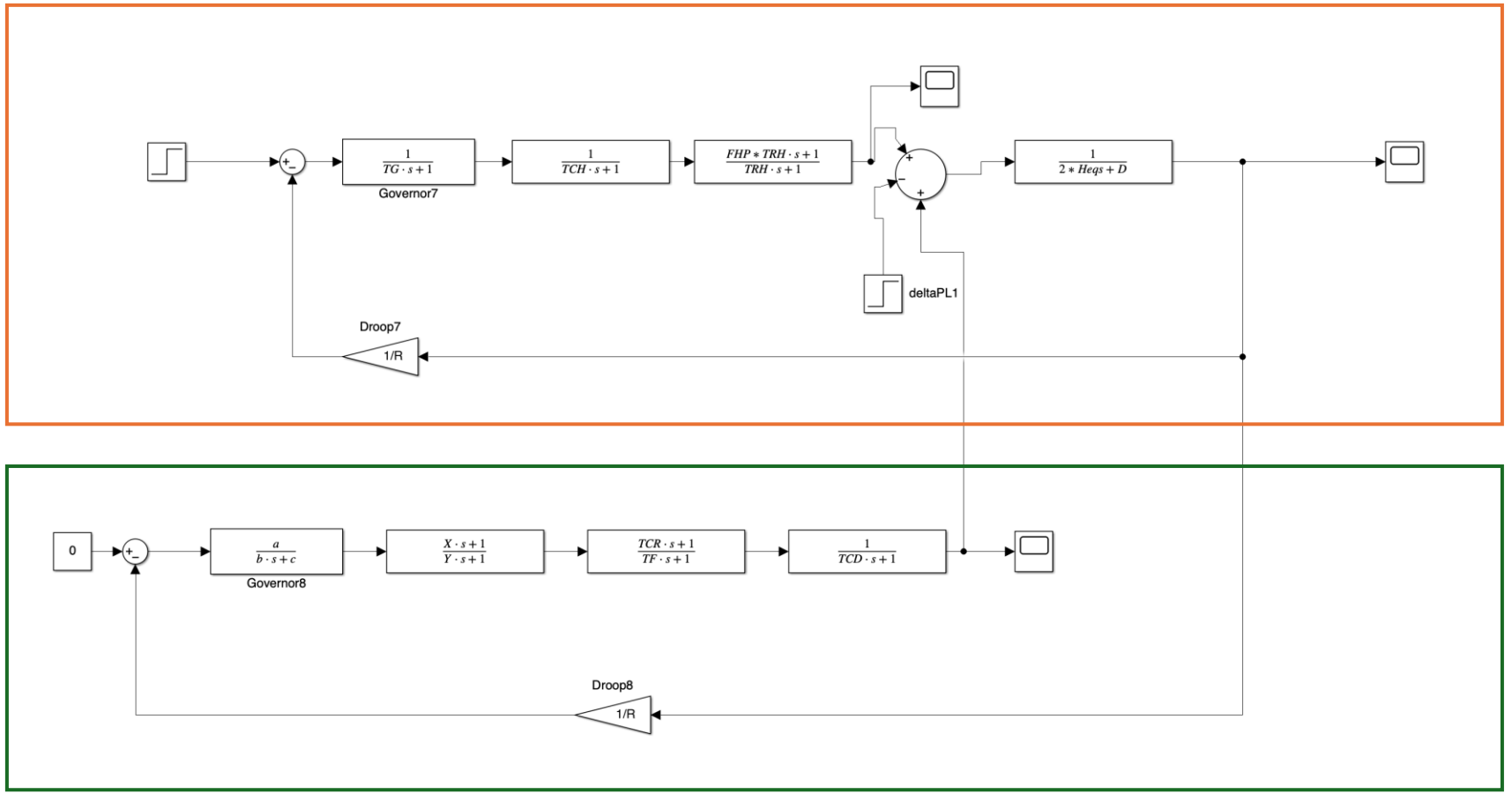
◀ Gas turbine

3. Compensator Design

Single-Reheat Steam Turbine

+

Gas Turbine



Method

3. Compensator Design



Single reheat steam turbine

Gain margin: 10.7db
Phase margin: 53.2deg



Gas turbine

Gain margin: 11.1db
Phase margin: 38.2deg



Single reheat + Gas turbine

Gain margin: 0.373db
Phase margin: 1.26deg

Method

3. Compensator Design



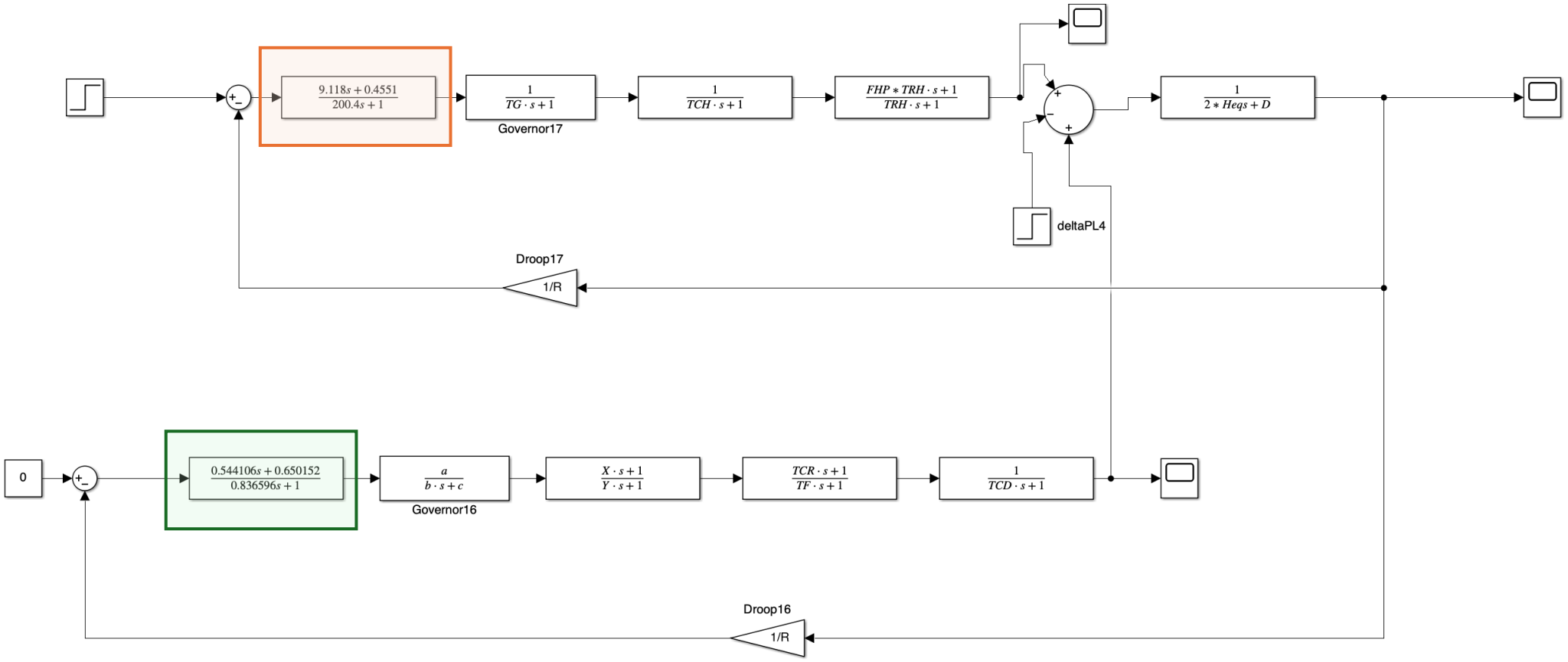
Single reheat + Gas turbine (with Step Recover)

» 시스템 안정성 저하
» 터빈 물리적 손상 위험
» 응답 시간 증가로 제어 성능 저하

Design Specifications
Phase Margin: 50°
Gain Margin: ≥6 dB 확보

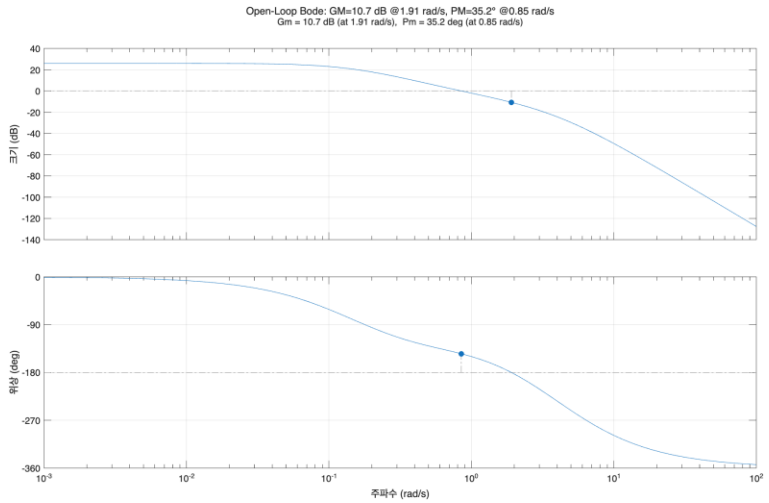
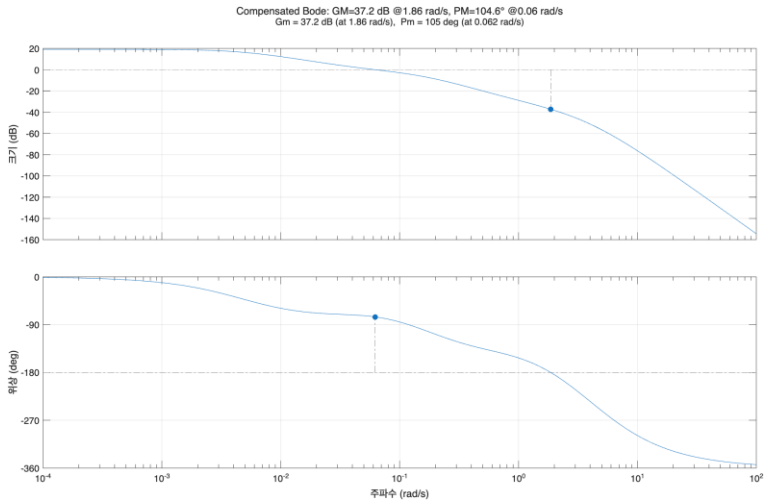
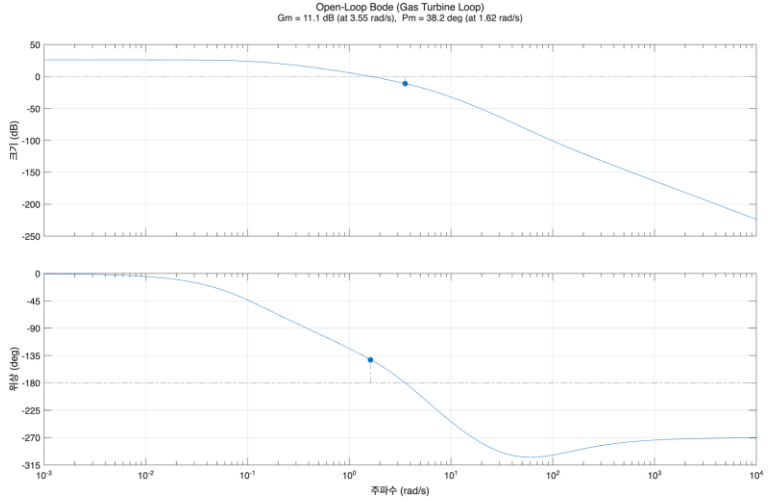
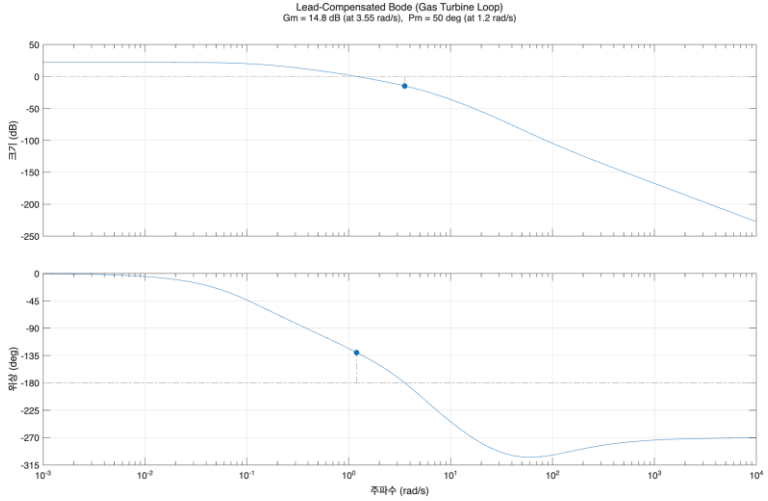
1st Specification Design

Individual compensators per turbine



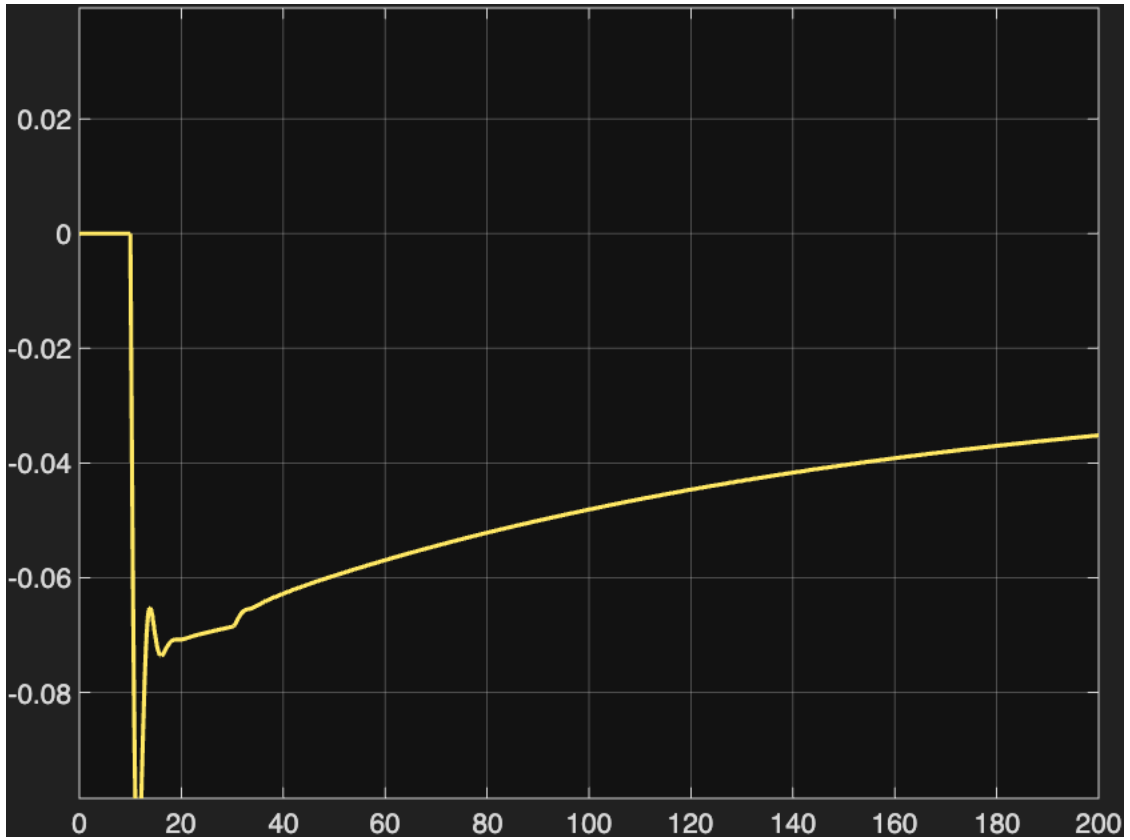
Method 3. Compensator Design

1st Specification Design

	Bode Plot	Bode Plot with Compensator
Single Reheat Steam Turbine		
Gas Turbine		

Method 3. Compensator Design

1st Specification Design



◀ Turbine model response

» Even if we make it stable, our goal is to **quickly adjust the response while reducing vibration**.

2nd Specification Design

1st Design Specifications
Phase Margin: 50°
Gain Margin: ≥6 dB 확보

→

2nd Design Specifications
Phase Margin: 40°–60°
Gain Margin: ≥6 dB 확보
ω_c: 기존 범위(1–4 rad/s) 근처 유지
응답 속도: 과도진동 억제, settling time 15–20 s 이내

» Design goal: quickly adjust the response while reducing vibration
» Maintained high crossover frequency (ω_c ≈ 1-4 rad/s) to widen bandwidth.→ Ensured quick response to input changes
» Phase Margin (PM) and Gain Margin (GM) requirements were specified as ranges (PM: 40°–60°) for practical balance

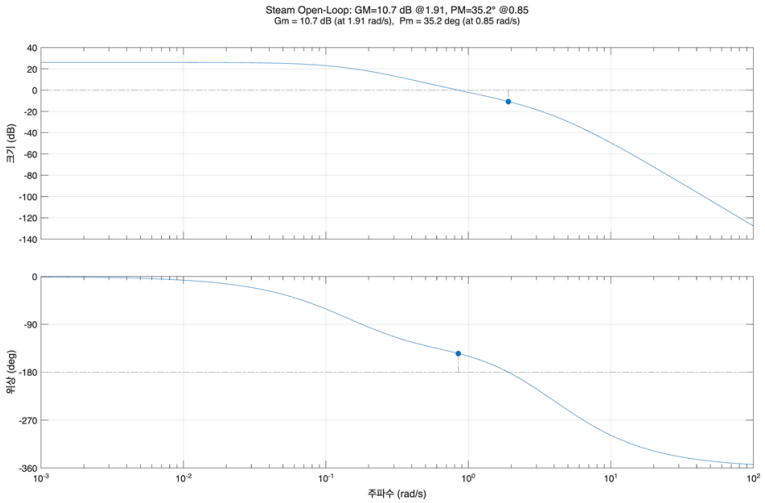
trade-off between **response speed** and **robustness**
→ Stability margins may naturally decrease when prioritizing fast response.

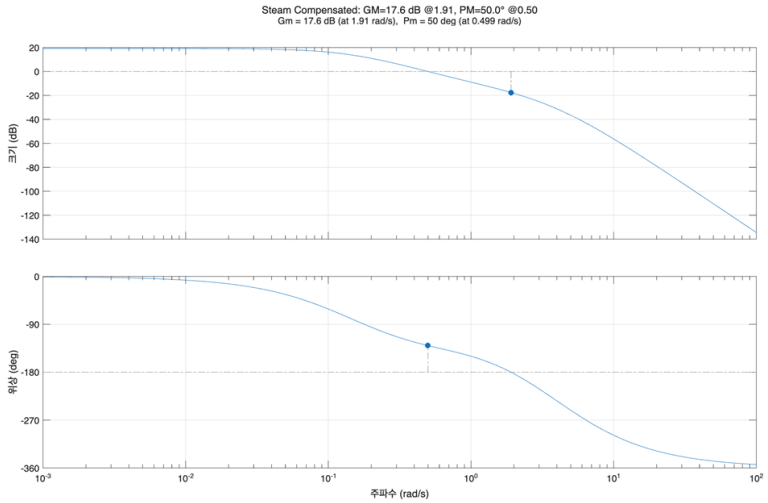
2nd Specification Design

Bode Plot

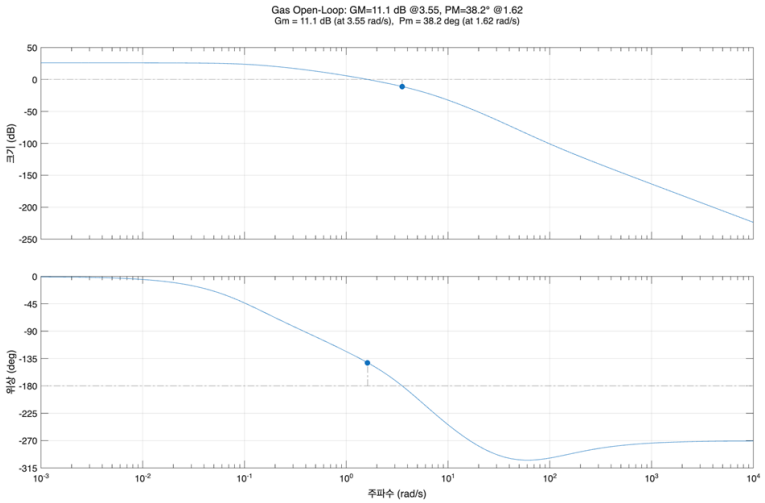
Bode Plot with Compensator

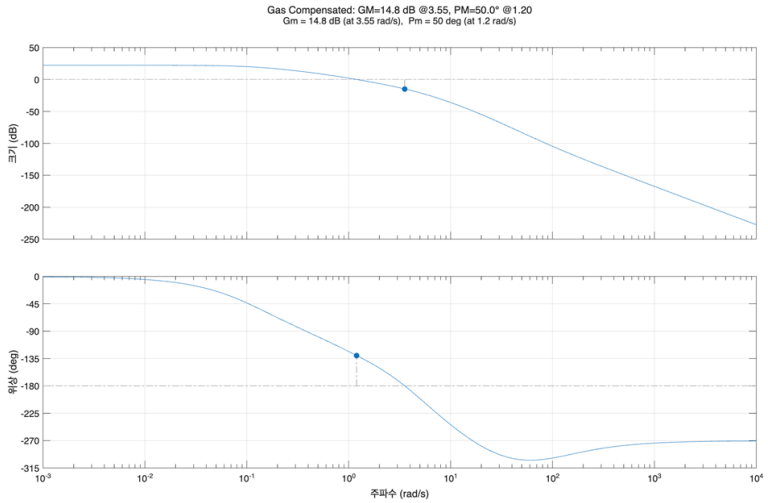
Single Reheat Steam Turbine



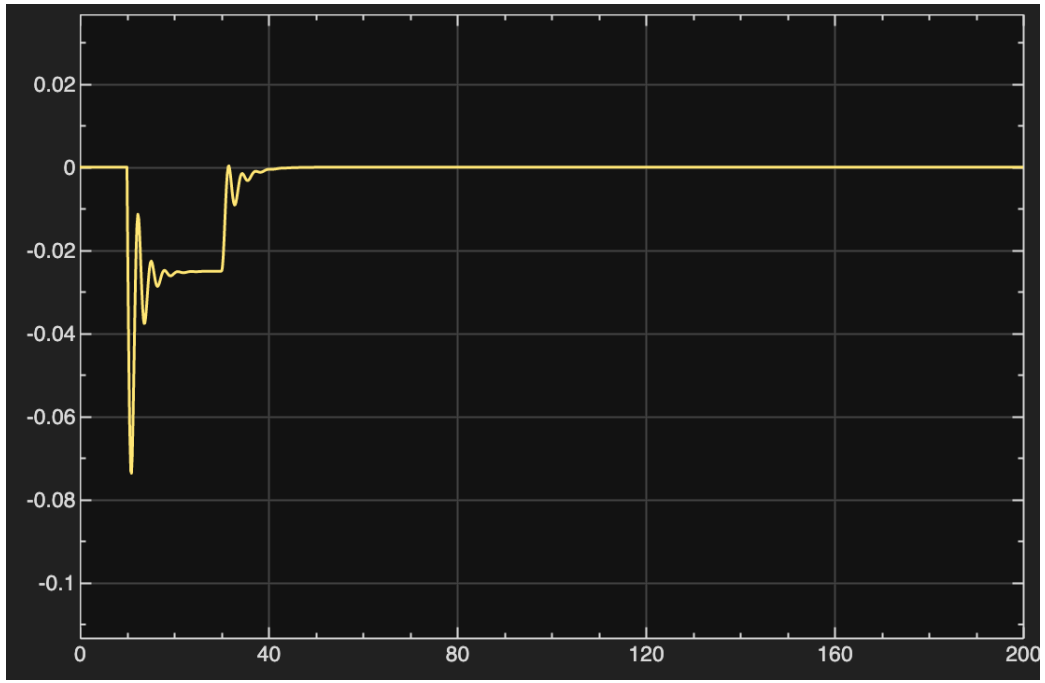


Gas Turbine

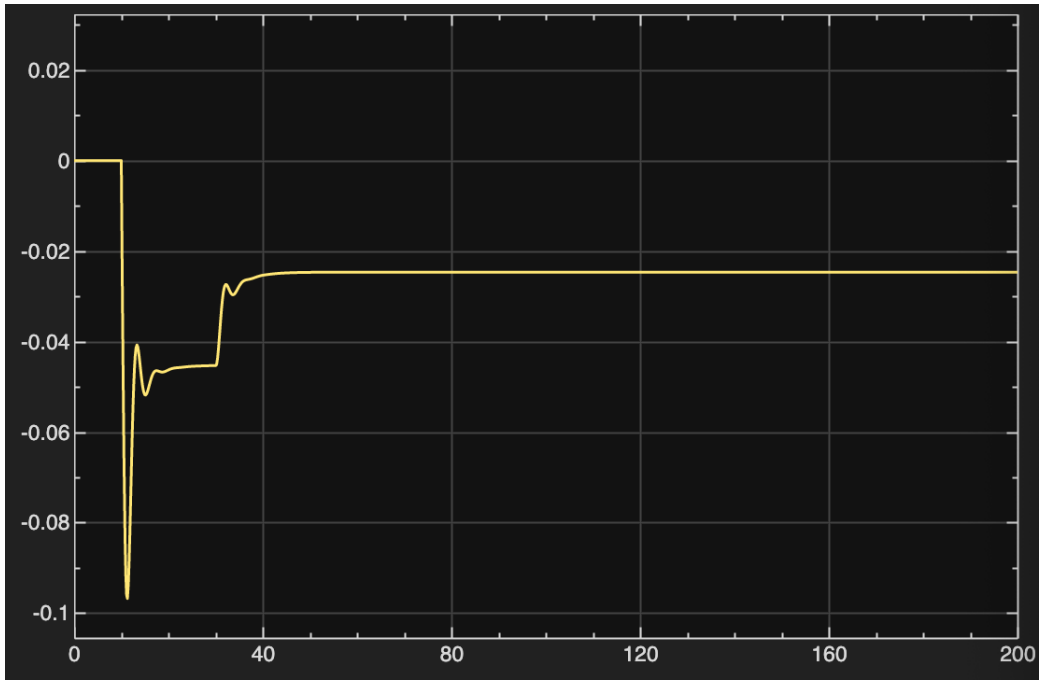




2nd Specification Design



Single reheat + Gas turbine
(without compensator)

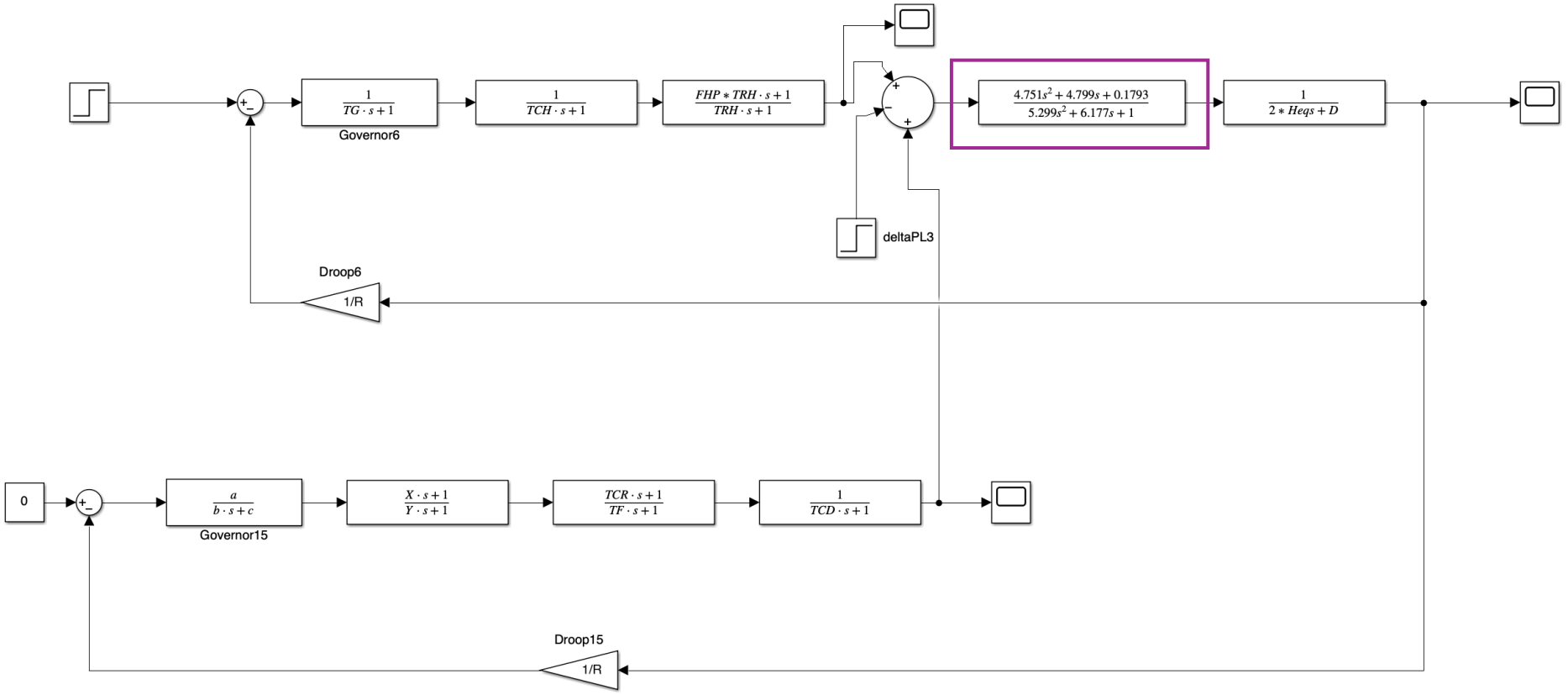


Single reheat + Gas turbine
(with compensator)

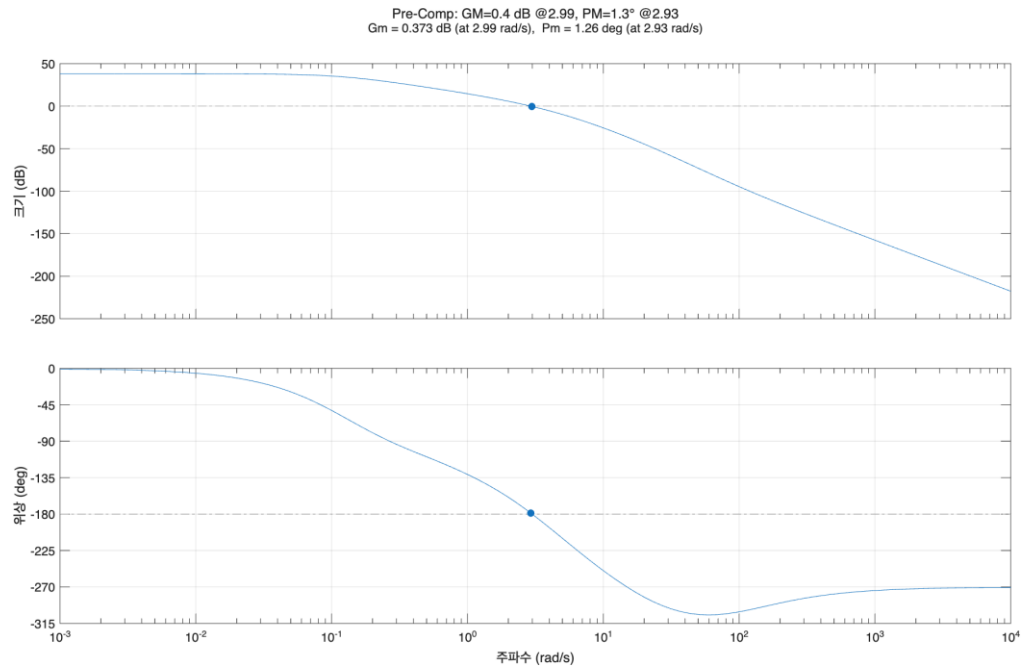
2nd Specification Design – Combined

: 하나의 보상기 설계

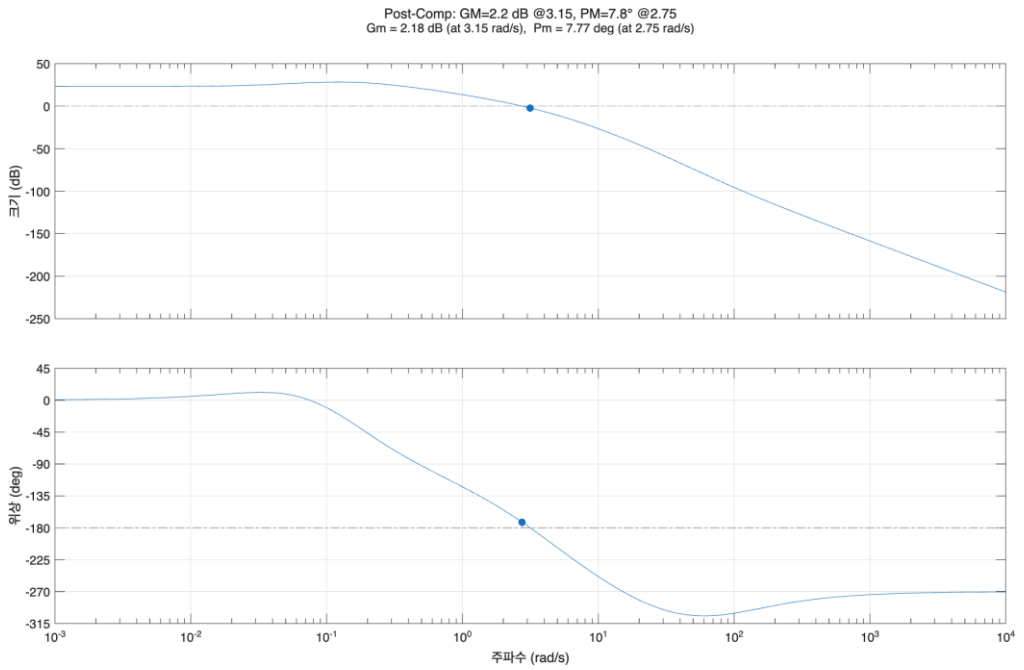
$$L_{\text{tot}}(s) = \frac{2}{R}\left[\frac{1}{(T_G s+1)(T_{CH} s+1)}\,\frac{FHP\cdot TRH\, s+1}{T_{CH} TRH\, s^2+(T_{CH}+TRH)s+1} + \frac{a}{b\,s+c}\,\frac{X\,s+1}{Y\,s+1}\,\frac{TCR\,s+1}{TF\,s+1}\,\frac{1}{TCD\,s+1}\right] \times \frac{1}{2\,H_{eq}\,s+D}.$$



2nd Specification Design – Combined



Bode Plot

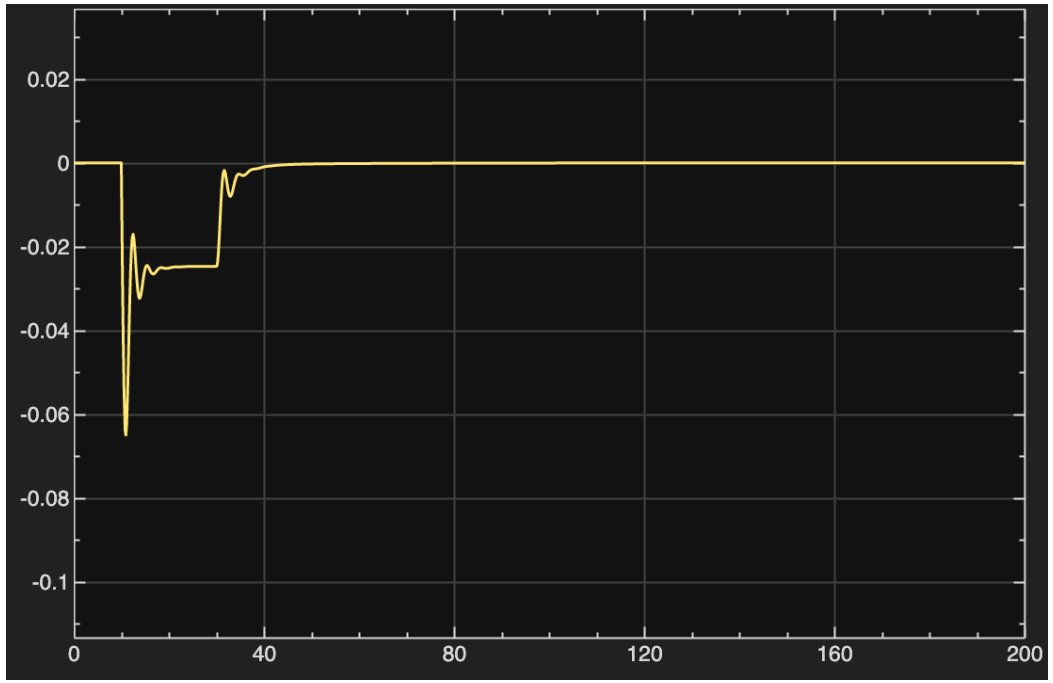


Bode Plot with Compensator

2nd Specification Design – Combined

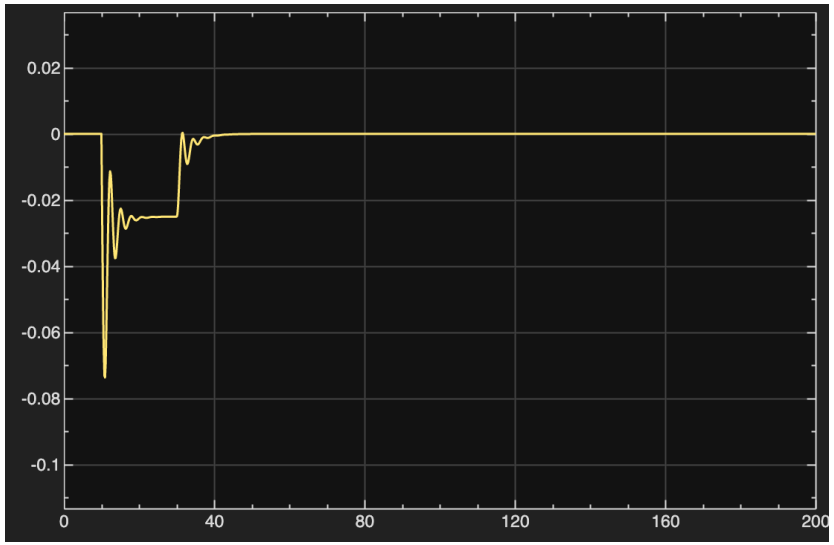


Single reheat + Gas turbine
(without compensator)

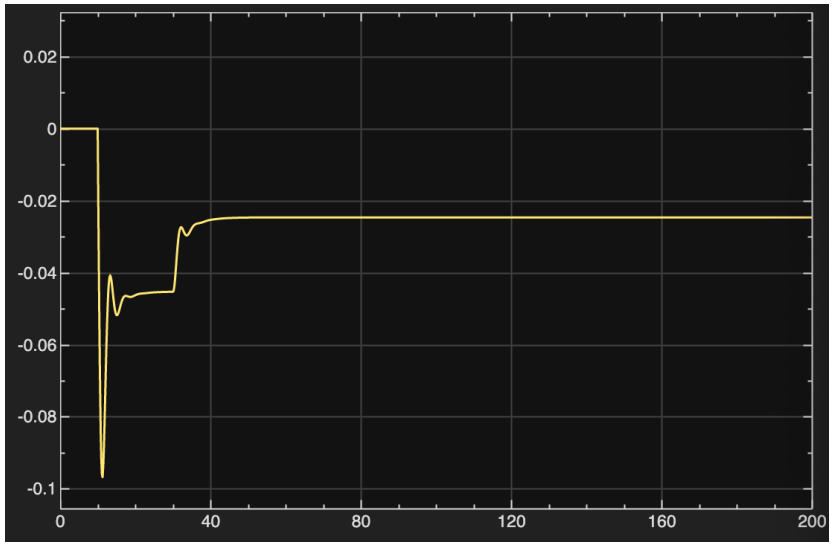


Single reheat + Gas turbine
(with compensator)

Limitation



(without compensator)

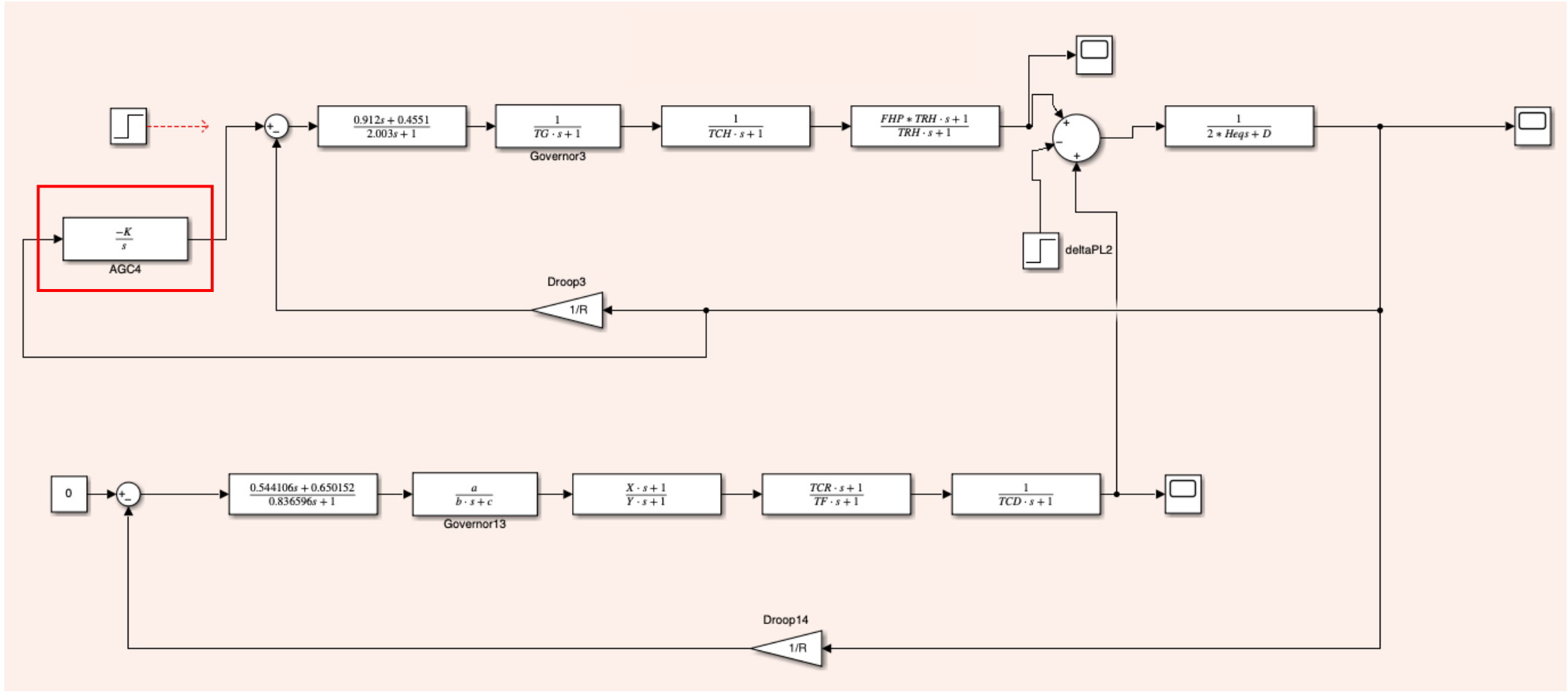


(with compensator)

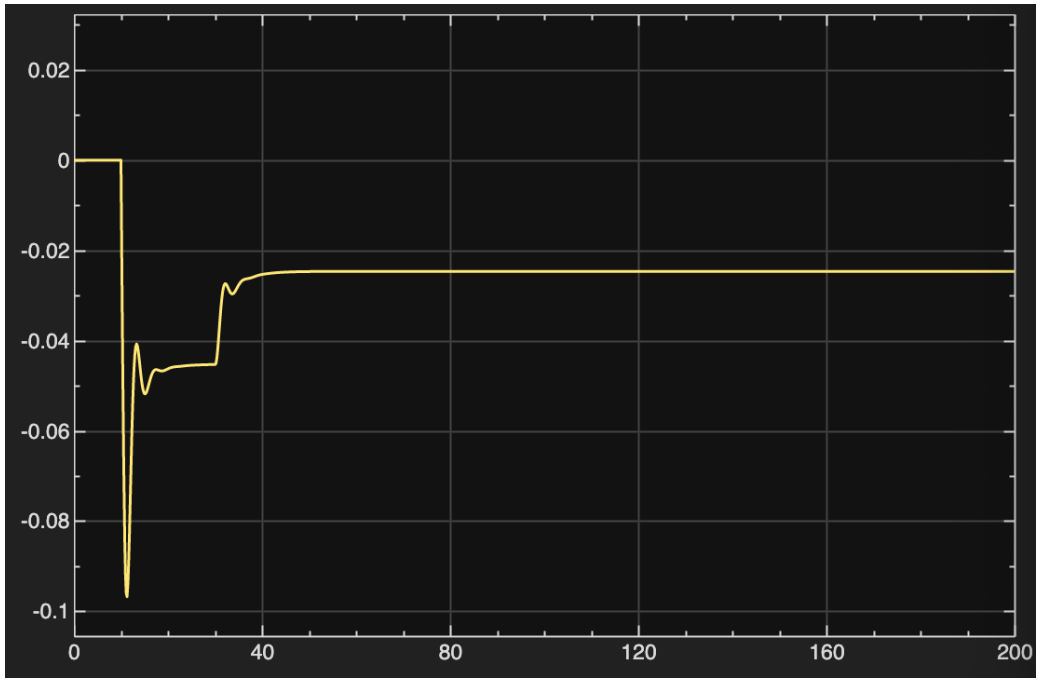
: 보상기를 설계 시 응동은 확실히 진동이 줄어드는데 step recover을 줘도 다시 0으로 올라오지 않음

→ Applying AGC

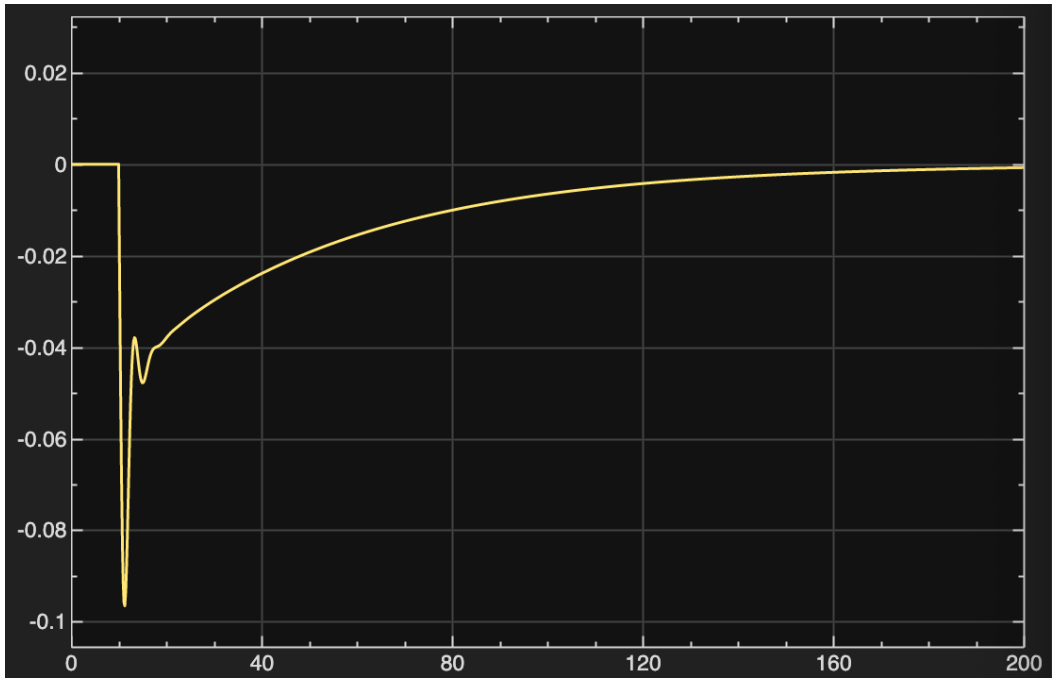
Extension


0.912s + 0.4551
2.003s + 1
Governor3
1
TG · s + 1
1
TCH · s + 1
FHP * TRH · s + 1
TRH · s + 1
1
2 * Heqs + D
deltaPL2
−K
s
AGC4
Droop3
1/R
0
0.544106s + 0.650152
0.836596s + 1
a
b · s + c
Governor13
X · s + 1
Y · s + 1
TCR · s + 1
TF · s + 1
1
TCD · s + 1
Droop14
1/R

Extension

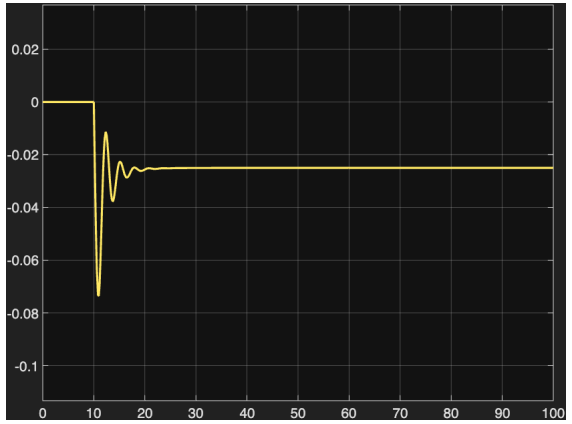


Without AGC



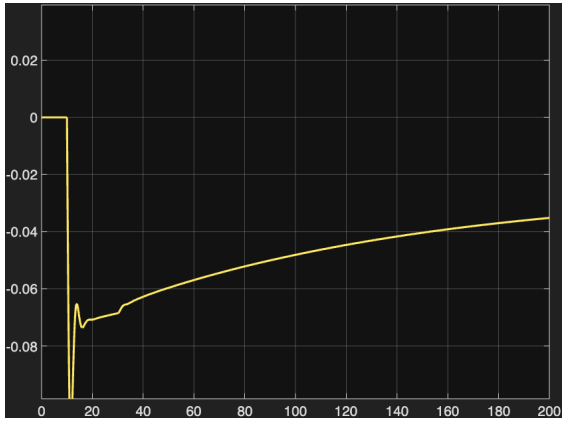
AGC

Conclusion



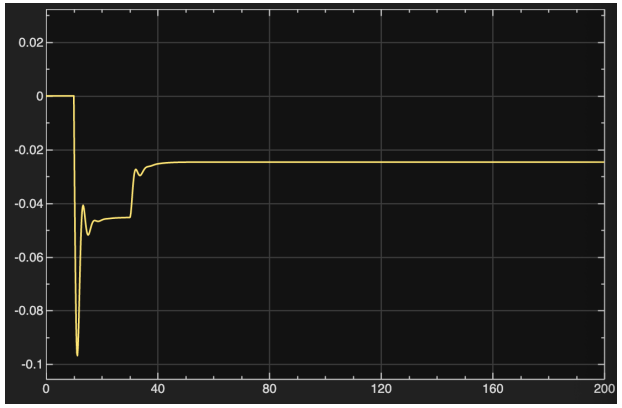
Single reheat + Gas turbine

Specification 1

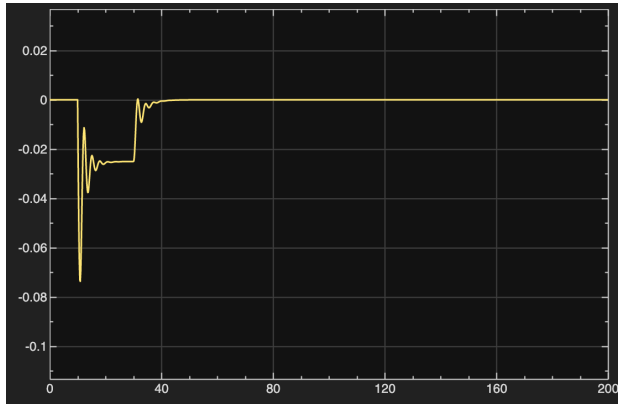


:phase margin improved

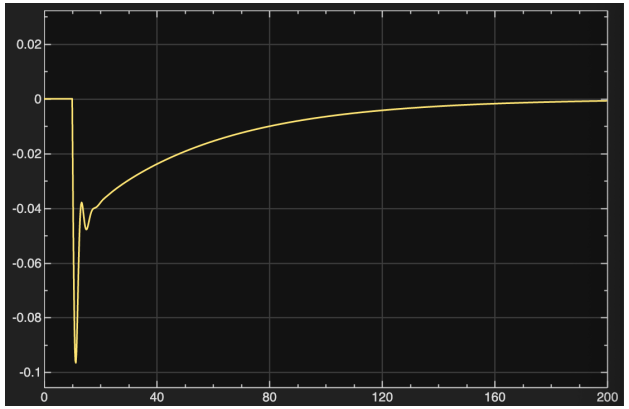
Specification 2



+ individual compensator



+ combined compensator



+individual +AGC